Demonstration Diversity Is Not Free at a Fifty-Episode Adaptation Budget

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Abstract

A practitioner adapting a pretrained vision-language-action model to a new embodiment has a fixed demonstration budget and must decide how to spend it. We pre-registered a comparison of four collection strategies at 50 demonstrations each, repetition (Clean), position diversity, recovery demonstrations and color diversity, holding the model (SmolVLA), task, hardware and training fixed: 600 rollouts across five evaluation cells and two seeds on a low-cost SO-ARM101. Both registered comparisons were null, one at a floor (0 of 120 held-out positions) and one at a ceiling. Trajectory telemetry explains both, and a third the success rates recorded but never explained: every other condition outscored the position-diverse policy by 2.4 to 4.6 times, yet that policy reached the lowest training loss of any run. Policies trained at a single position execute a fixed sweep that stops covering the cube within one inch. The position-diverse policy aims at the cube inside its training hull, yet never leaves its home pose in a fifth of episodes. A follow-up policy trained at two positions with 25 demonstrations each executes perfectly at both and, everywhere else, returns to one of its two trained bearings: density buys execution, not interpolation. Policies also inherit their demonstrator: a demonstrated mid-carry drop is reproduced within one degree of its location, and demonstration velocity orders completion time. Success rate alone would have reported no effect anywhere, which is the practical hazard at this end of the ecosystem.

1 Introduction

Vision-language-action (VLA) models are deployed through a common recipe: pretrain on large corpora, then adapt to a specific embodiment and task with a small set of teleoperated demonstrations. At the accessible end of the ecosystem that set is a few dozen episodes collected in an afternoon, and every practitioner who collects one faces the same decision with little evidence: spend a fixed budget on repetition of one configuration, or on diversity, and if diversity, along which axis? Results at large scale favor diversity, but the recommended rate is out of reach here: Lin et al. (2024) report plateaus only once each environment-object pair carries roughly 50 demonstrations, and a 50-episode budget spread over ten positions samples each at five. At this scale diversity is not a choice of axis but a decision to sample every configuration below the recommended density. Whether that guidance survives at the budgets low-cost adapters actually use is untested, and it is the focus of this paper.

Holding the model (SmolVLA), task ("Pick up the cube and place it in the cup"), hardware, budget (50 demonstrations) and training configuration fixed, we vary only the collection strategy: a repetition baseline and three conditions that each differ from it in exactly one factor, cube position, cube color, or demonstrated failure recovery. The protocol was pre-registered with dated amendments, two matched comparisons and all metrics were committed before collection, and every policy was evaluated on five cells at two training seeds, 600 registered rollouts and 107 exploratory ones, with labels validated against telemetry.

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Both registered comparisons came back null, and the pattern of the nulls is the finding: every policy scored zero on held-out positions, 120 episodes without one success, and both policies compared on a held-out color sat at ceiling. Success rate, the field's default currency, reported no effect anywhere. The recorded trajectories reported four. Policies trained at one position command the same fixed sweep regardless of the scene, and that sweep stops covering the cube within one inch, which is why the position cell is a floor rather than a gradient. The position-diverse policy is the only one whose aim tracks the cube, and it still posts the lowest grid total in the study, failing to leave its home pose in roughly a fifth of its episodes while reaching the lowest training loss of any run. A follow-up policy trained on two positions at 25 demonstrations each executes perfectly at both, but at held-out positions it returns to one of its two trained bearings. Coverage and competence are bought separately: spreading a fixed budget across more configurations buys aim and loses execution, and concentrating it buys execution and no interpolation. Finally, the policies inherit their demonstrator: a deliberate mid-carry drop is reproduced within a degree of its demonstrated location while the recovery that followed it rarely transfers, and demonstration velocity orders completion time. Linear probes locate all four in the readout rather than the representation: cube position is decodable from every policy, including those that ignore it, and outcome is not decodable before the grasp. We argue for two changes of practice at small budgets: **(1) price diversity rather than assuming it**, since every configuration added is demonstrations removed from every other, and below some density per configuration the policy stops executing at all; and **(2) evaluate with trajectory-level measurement and pilot-tested cell difficulty**, since a policy cleared on success rate alone can carry an inherited failure or a blindness to the scene into deployment.

2 Related work

VLA models and small-budget adaptation. VLA models map camera images and a language instruction to robot actions, trained end to end from demonstrations (Brohan et al., 2022, 2023; Octo Model Team, 2024; Kim et al., 2024; Black et al., 2024); SmolVLA (Shukor et al., 2025) is a compact VLA designed to be fine-tuned and run on consumer hardware. The open models in this line share a deployment story in which a pretrained policy is adapted to a new embodiment with tens to low hundreds of teleoperated episodes (Octo Model Team, 2024; Kim et al., 2024; Black et al., 2024; Shukor et al., 2025), by behavior cloning over predicted action sequences (Zhao et al., 2023; Chi et al., 2023). That adaptation step is our subject.

Demonstration quantity, diversity and quality. At scale, diversity helps: Lin et al. (2024) find power-law improvements with the number of environments and objects, and Xie et al. (2023) decompose the generalization gap by factor, finding camera position hardest and background easiest, with texture factors close to the spatial ones in difficulty. Our spatial cell reproduces their ordering at its extreme; our appearance cell does not, for a reason Section 6 gives. They randomize object position rather than evaluating it, since it is hard to place precisely on hardware; we hold it to marked positions and measure it, and their finding that most factors do not compound, 16 of 21 pairs within six percent, is the premise of our one-factor-per-cell design. A complementary line examines quality: demonstrator skill and heterogeneity strongly affect cloning outcomes (Mandlekar et al., 2021), and Belkhale et al. (2023) argue that state coverage bought at the expense of action consistency can yield a worse policy than narrower data. A fixed 50-episode budget forces that trade; we measure what it costs and what the failure looks like.

Behavior cloning pathologies. Behavior cloning fails in ways success rate does not name. One family is distribution shift, where the policy's own errors compound (Ross et al., 2011; Laskey et al., 2017). Another, closer to our findings, is that the policy copies more than the task: causal confusion (de Haan et al., 2019), the copycat problem (Wen et al., 2020), and shortcut learning generally (Geirhos et al., 2020). We add two instances measured on hardware, a demonstrated failure reproduced at its location without its correction, and a demonstration-velocity signature that transfers from teleoperator to policy, neither of which success rate distinguishes from ordinary failure.

Evaluating robot policies. Kress-Gazit et al. (2024) treat policy evaluation as an empirical science, Snyder et al. (2025) give near-optimal stopping rules at realistic trial counts, and distributed or simulator-assisted schemes attack the cost of real evaluation directly (Atreya et al., 2025; Li et al., 2024). We adopt this posture: pre-registration with dated amendments, metrics fixed before collection,

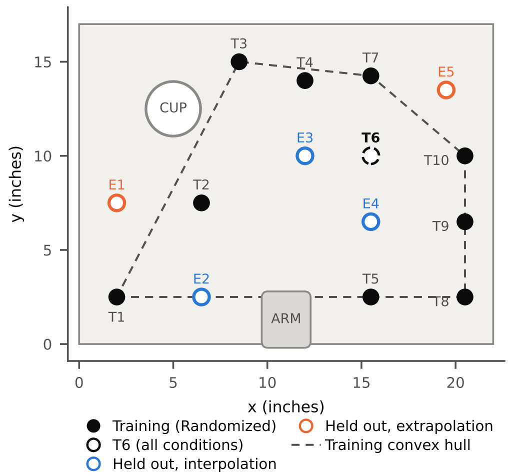

Figure 1: Cube starting positions in the overhead frame. Randomized trains on T1 to T10, every other registered condition on T6 only; the exploratory Density policy trains on T6 and T2. Held-out E1 and E5 lie outside the training convex hull.

Wilson and Newcombe intervals, a power statement and telemetry-validated labels. Our two nulls, one floor and one ceiling, are a concrete argument for piloting cell difficulty before registering a comparison, a practice we did not follow and now recommend.

Probing and failure detection. Linear probes read information from intermediate representations (Alain and Bengio, 2016); for VLAs, SAFE (Gu et al., 2025) detects failures of generalist policies from internal features. We use probes in a narrower role: to test whether information a policy demonstrably does not use is nonetheless linearly present.

3 Setup

Apparatus and task. An SO-ARM101 arm with a leader for teleoperation and a follower for rollout; two cameras, overhead and wrist, 640×480 at 30 fps; the fixed instruction "Pick up the cube and place it in the cup." Camera position, lighting and home pose are locked for the study. Objects are 1-inch foam cubes identical across colors. Appendix A gives geometry, lighting and photographs.

Collection conditions. Four conditions of 50 demonstrations each: Clean, and three that differ from it in exactly one factor. **Clean:** a red cube at position T6, every demonstration a first-try success. **Randomized:** only start position varies, ten marked positions including T6, cycled in five passes (Figure 1). **Recovery:** 20 of 50 demonstrations include a deliberate mid-carry release from 4 to 5 inches, a re-grasp and completion, in fixed slots. **Color-varied:** five cube colors for ten demonstrations each, green held out. Varied conditions cycle rather than block, so teleoperator drift cannot masquerade as a factor effect. The budget matches Lin et al. (2024)'s recommendation of 50 demonstrations per environment-object pair: Clean is sampled at that rate, Randomized at a tenth of it. Demonstration pace was uncontrolled (16.6 to 21.4 s), and because the budget is fixed in episodes, conditions receive 9.9 to 12.9 passes over their own data; both are covariates (Appendix I). Color-varied was re-collected before training: the first collection tried to match Clean's episode length, came out 39 % slower than the retained set, and is reused as the pace probe.

Training. Every policy is fine-tuned from `lerobot/smolvla_base` with identical hyperparameters (batch 32, 10,000 steps, single A100; Appendix B), and the final checkpoint is always evaluated. LeRobot's default cosine schedule is configured for 30,000 steps against our 10,000, so the evaluated checkpoint sits near 79 % of peak learning rate rather than fully annealed; this applies identically to every run and is disclosed rather than corrected, since changing it afterwards would break the comparison the fixed configuration protects. Only the action expert is trained: LeRobot's defaults freeze the vision encoder and language backbone, leaving roughly 100M of 450M parameters trainable, so all policies share an identical perceptual front end and every behavioral difference is attributable to the action expert. The training seed is the only setting varied between replications: the full grid runs at seed 1000 and is replicated at seed 2000, and conditions are never compared across seeds. The configurations differ only in dataset, seed, output path and bookkeeping fields, and are released with the code.

Evaluation. Each policy is evaluated on five cells of 15 scored episodes (16 recorded, index 0 a pre-committed warmup). **In-Distribution:** red cube at T6. **New Positions:** five held-out positions, three episodes each, split by the training convex hull. **Reduced Lighting:** one LED off, a 30 % drop in image brightness. **Different Object:** a green cube at T6. **Distractors:** four objects at marked positions, three of them Randomized training positions, deliberately adversarial for that condition. An episode succeeds if the cube rests in the upright cup within 45 s; the criterion is binary and stricter than the base model's reference evaluation, which awards half credit for a grasp. The policy emits 50 actions per forward pass and executes all of them before the next, an open-loop chunk rather than a receding horizon, one decision every 1.67 s. The registered grid is 4 conditions $\times$ 5 cells $\times$ 15 episodes $\times$ 2 seeds, 600 rollouts.

Scoring, validation and instruments. Episodes are scored live with all video retained; ambiguous episodes are re-scored from video, and failure modes are coded against a fixed ten-term vocabulary (Appendix G). As a single-operator study, demonstrator and evaluator roles were not separated and scoring was not blind to condition; mitigations are a binary criterion fixed before collection, a pre-registered cell order, and three label families validated against telemetry including a full audit of every scored episode against outcome-independent screens, which found one transposed pair, corrected with no change to any statistic (Appendix C). Trajectory analysis maps base rotation to workspace bearing through a linear fit on the 50 Randomized training grasps, validated at two locations it never saw (leave-one-out median error 0.86 degrees); radial accuracy is insufficient, so claims are stated in bearing, from commanded joint positions. These analyses are post hoc and exploratory.

Pre-registration and statistics. The protocol was pre-registered before any data was collected and has 30 dated amendments (Appendix H). Registered metrics are per-cell success with Wilson 95 % intervals, the generalization gap, and two matched comparisons, Randomized against Clean on New Positions and Color-varied against Clean on Different Object, with Newcombe intervals and Fisher tests. Seeds are never pooled. Intervals are roughly ± 14 points at $n = 75$, so differences under about 13 points are unresolvable and nulls are reported as inconclusive. Episodes within a cell share a scene, so pooled 75-episode intervals are narrower than a cluster-corrected version; per-cell intervals at $n = 15$ are unaffected, and the pooled comparison bounds seed sensitivity rather than testing it. Three exploratory probes, each declared before it ran, extend the grid: a displacement probe (Clean with the cube 1 and 2 inches from T6, 8 episodes per point at each seed), a pace probe (the slow-pace Color policy), and a density probe (Section 5.4).

4 Both registered comparisons are null

Table 1 reports the grid. On New Positions no policy succeeded in any of the 120 held-out episodes, so the cell cannot distinguish conditions in principle. On Different Object, Color scores 15/15 at both seeds and Clean 15/15 and 12/15; the one contrast with room to move goes in the predicted direction ($+0.20$, Fisher $p = 0.224$). Both matched comparisons are therefore null, one at a floor and one at a ceiling (Appendix E, Table 2). The pre-committed generalization gap is bounded above by the in-distribution rate (Appendix E); held-out means are the interpretable comparison, 0.667 and 0.517 for Clean, 0.667 and 0.733 for Color, 0.517 and 0.367 for Recovery, 0.100 and 0.117 for Randomized.

What the grid does record is the Randomized collapse: 12/75 and 13/75 against Clean's 55/75 and 46/75, with identical in-distribution and reduced-lighting counts at both seeds, while the three single-position conditions score $2.4\times$ to $4.6\times$ higher. Training loss does not predict it: Randomized reaches the lowest final loss of any run, 0.038 and 0.036, and the worst success rate in the study (Figure 2; Figure 11, Appendix B; Table 7, Appendix F), consistent with the difficulty of offline policy selection reported by Mandlekar et al. (2021). The collapse is not a training failure. Section 5 traces it, and both nulls, to measured mechanisms.

Table 1: Successes out of 15 rollouts per policy and cell, both seeds. New Positions is 0/15 for every policy at both seeds, so the registered comparison on that axis sits on a floor.

Condition	Seed	In dist.	Reduced light	Diff. object	Distractors	New positions	Total
Clean	1000	15	12	15	13	0	55/75
Color	1000	13	12	15	13	0	53/75
Recovery	1000	8	7	11	13	0	39/75
Randomized	1000	6	5	0	1	0	12/75
Clean	2000	15	9	12	10	0	46/75
Color	2000	12	14	15	15	0	56/75
Recovery	2000	9	5	5	12	0	31/75
Randomized	2000	6	5	2	0	0	13/75

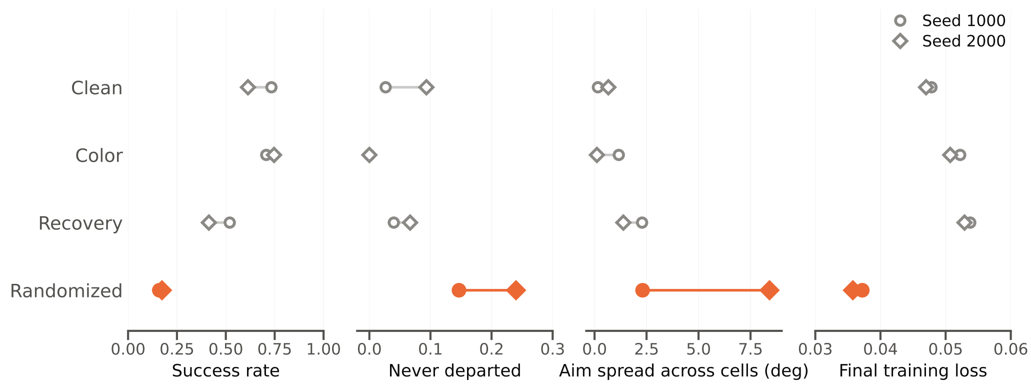

Figure 2: **What success rate alone would miss.** Per condition, circles seed 1000 and diamonds seed 2000: grid success rate, fraction of episodes in which the arm never left the home pose, spread of median commanded bearing across the four same-target cells, and final training loss. Randomized is last on the first, worst on the second and third, and best on the fourth.

5 Telemetry explains both nulls and the collapse

All analyses in this section are exploratory. Video of each behavior, one episode per finding, is in the repository's `media/` directory and README.

5.1 Six of eight policies aim at the trained position regardless of the cube

We compare the median commanded bearing of the base joint across cells and policies; the base joint is used because its angle is tightly determined by cube position in the demonstrations while elbow and wrist vary widely (Figure 3, left). For every policy except Randomized the commanded bearing sits within about two degrees of the required bearing for T6 in every cell, including New Positions, where the cube is somewhere else. Only Randomized's aim moves with the cube, by an order of magnitude more than any other policy's cross-cell spread, and it is also the only policy whose aim is pulled by distractors: 8.4 degrees of spread across same-target cells at seed 2000 and 2.3 at seed 1000, against at most 2.3 for the other conditions. The policy that saw ten cube positions is the only one diverted by objects that are not the cube.

5.2 The fixed sweep stops covering the cube within one inch

Given that every single-position policy aimed at T6 regardless of the scene, we measured how far the cube can move before Clean's motion stops covering it, the continuous-radius experiment of Xie et al. (2023) applied to a factor they randomized rather than measured. Figure 3 (right) plots the margin between the furthest bearing the arm reaches in an episode and the bearing the cube requires. At T6 the sweep clears the requirement by 2.8 degrees at both seeds and succeeds 15/15. At one inch the

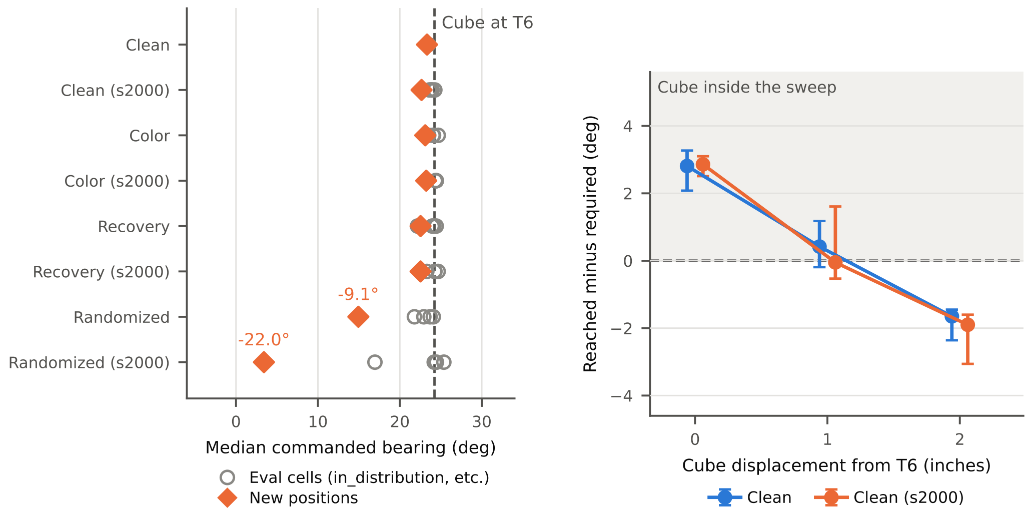

Figure 3: **Left: only Randomized's aim moves when the cube does.** Median commanded bearing per policy; the dashed line marks what a cube at T6 demands. **Right: the sweep stops covering the cube past one inch.** Labels give the episodes reaching the required bearing.

margin falls to 0.4 degrees and to zero, and the sweep clears the bearing the cube's center requires in 6 of 8 and 4 of 8 episodes. The cube subtends about ± 2.9 degrees at that distance, so episodes short of center can still reach its near face: contact, scored at the bench rather than from telemetry, occurs in 6 of 8 and 8 of 8, and success in 0 of 8 and 3 of 8. At two inches the sweep falls 1.7 and 1.9 degrees short, contact occurs in 0 of 8 and 1 of 8 episodes, and success is 0/8 at both; the failure mode shifts from contact without a grasp at one inch to no contact at two (Table 6). The one-inch contacts are the fixed sweep happening to pass over the displaced cube rather than correction: displacing the cube by 5.13 degrees of bearing moves the commanded trajectory by 0.67 and 0.37 degrees. The policy is not failing to localize the cube; it is not looking.

5.3 Randomized aims at the cube, and often does not move at all

Randomized's aim tracks the cube, and how well depends on where the cube sits relative to the convex hull of its ten training positions: E2, E3 and E4 are interpolation, E1 and E5 extrapolation. Across the 12 held-out episodes in which the gripper closed (Figure 4, left), median absolute bearing error is 2.1 degrees at the seven interpolated grasps, well under a cube width, and 31.1 degrees at the five extrapolated ones, whose signed errors ($+38.8$, $+51.6$, -6.5, -16.2, -31.1) all point toward the center of the training distribution (Mann-Whitney $U = 34$, $p = 0.0025$). In the other 18 held-out episodes the gripper never closed.

Position variation therefore bought a policy that localizes the cube inside its training hull and converted none of it into success. The execution failure has a specific form: Randomized never left the home pose in 29 of 150 episodes, 11 at seed 1000 and 18 at seed 2000, against 11 of 182 for Clean, 4 and 7 per seed (Table 8). Latency does not separate the conditions the same way, since Clean is also slow to depart; the discriminating statistic is whether the arm departs at all. That is not a worse reach but a failure to commit to one: fifty demonstrations across ten positions left too few at each to specify an action confidently.

5.4 Twenty-five demonstrations per position buy execution, not interpolation

One exploratory policy, declared before collection, asks whether that cost is demonstration density rather than diversity as such. **Density** spends 25 demonstrations at T6 and 25 at T2, five times Randomized's rate per position. T2 sits on the opposite side of the arm (-31 against $+24$ degrees of base bearing), so no fixed sweep can succeed at both and the policy must condition on the image to choose a direction. It is evaluated at T6, at T2 and on the registered New Positions cell; single seed,

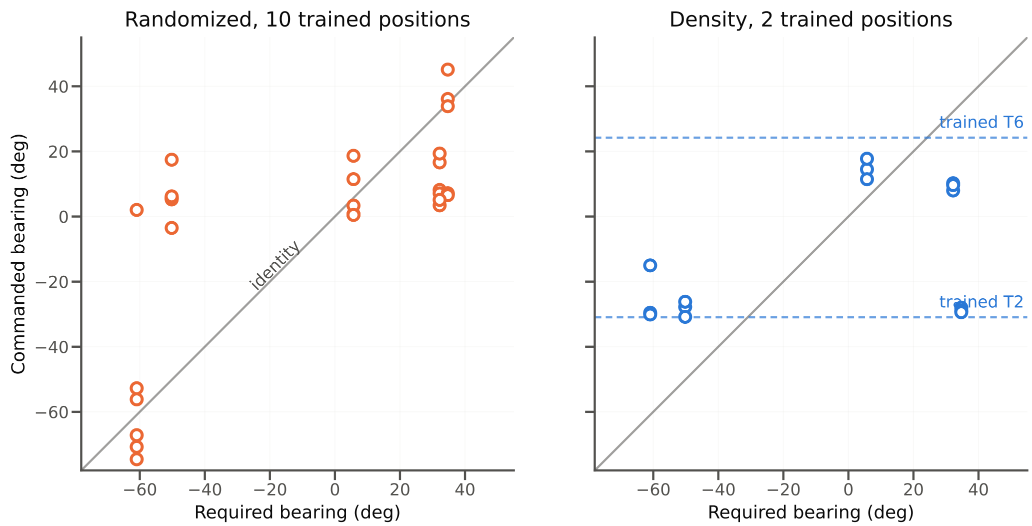

Figure 4: **Two ways to fail a held-out position.** Settled commanded bearing (last quarter of the episode) against required, one point per episode. Left: Randomized tracks the cube where it interpolates and loses it where it extrapolates. Right: Density returns to one of its two trained bearings (dashed) whatever the cube requires; the T6-side clusters sit below the line because of late-episode drift toward the middle.

recorded two weeks later by the same operator, by then more practiced, and never pooled into the grid.

It succeeded 15/15 at both trained positions and never once failed to depart, with the shortest departure latency (0.5 and 0.6 s) and fastest motion of any policy in the study, against Randomized's 6/15 at T6 and 29 no-departure episodes in 150. Grasp bearing at the two positions is 24.1 and −32.6 degrees against 24.2 and −31.0 required. The collapse was a density effect, and the floor price lies between 5 and 25 demonstrations per position. What density did not buy is generalization: 0/15 on New Positions, and the failure has a third form (Figure 4, right). At E1, E2 and E4 the arm settled at T2, −27.8, −29.6 and −28.9 degrees against −31.0 trained and 22 to 64 degrees from the cube; at E4 it first moved toward T6, on the cube's side of the arm, then crossed to T2. At E3 and E5 it went to T6 and drifted toward the middle late in the episode, settling at 14.5 and 9.6 degrees against 24.2 trained and 5.7 and 32.2 required. At four of five positions the settled bearing is closer to a trained bearing than to the cube's; E3, near the midpoint of the two, is the exception. The policy learned two behaviors and a selector that conditions on the scene, not a position map, and it did so at the highest final loss of the ten runs. Ten positions at five demonstrations produced a policy that localizes the cube and cannot act; two at twenty-five produced one that acts and cannot localize. The two settings sit at opposite ends of that trade; whether an intermediate split buys both is untested.

5.5 Position is in the representation either way, and outcome is not

We probe the action expert's final hidden state with two linear readouts (Appendix D). Cube bearing is decodable from every policy at both seeds, median error 8.6 to 11.4 degrees against a 42-degree chance floor, with differences between policies inside their intervals (Figure 13); holding out whole positions instead of episodes collapses decoding to 26 to 42 degrees, so the code is interpolative. The probe reads the only part of the network fine-tuning modifies, while the backbone supplying it is frozen and shared; the position code it traces back to comes from SmolVLA's pretraining on 481 community datasets, not from the 50 demonstrations. Position therefore survives into the trained module of every policy, including those whose commanded bearing never moves. What collection strategy changes is not what the network can see, but what the action output is conditioned on.

The outcome probe asks whether the representation forecasts success early. It does not. Sweeping an information cutoff across the episode in the six policies with enough outcome variance, first-window

AUC spans 0.53 to 0.74 and clears a within-cell permutation null in one of six policies at $p = 0.046$, which does not survive correction for six tests ($\alpha = 0.0083$) and does not replicate at the other seed. Decodability rises as the grasp unfolds, reaching 0.74 to 0.98 by ten seconds, and a full-episode control reaches 0.99 to 1.00 (Figure 14). This is consistent with SAFE (Gu et al., 2025), which detects failure from VLA features as it begins to unfold; in a short single-grasp task, "as it unfolds" and "at the grasp" coincide, leaving no pre-contact window in which a forecast could exist.

5.6 Recovery reproduces the demonstrated failure, not the correction

The Recovery policy releases the cube mid-carry at a median bearing of -11.7 degrees against the demonstrations' -10.8, one degree apart on an instrument accurate to 0.86 degrees, both about 65 % of the way from cube to cup (Figure 12). The release detector cannot see drops within 10 degrees of the grasp, which biases both estimates toward the cup equally. The correction did not transfer at the same rate: 6 of 53 releases were followed by the re-grasp that completes every demonstration, all six at seed 2000.

5.7 The demonstrator's tempo transfers

Completion time on successes tracks demonstration velocity. The retained Color dataset was the fastest collected, 16.6 s per demonstration, and its policies finish in 13.4 and 13.5 s; the slow-pace policy, trained on the same task and colors demonstrated at 23.0 s, finishes in 22.4 s, and two of its failures grasped the cube and ran out the window before delivery. Demonstration velocity in degrees per step and mean completion time are perfectly rank-ordered across the five datasets (Spearman $\rho = -1.0$, $n = 5$, $p = 0.017$, the floor at this n), and the four with policies at both seeds reproduce the ordering; velocity rather than seconds per demonstration is the ordering quantity, since Recovery is the slowest dataset to record and the third fastest to execute. Four of the five differ in content as well as pace, so the slow-pace pair is the controlled comparison; the slow-pace dataset was not a pre-registered condition and is analyzed as exploratory.

6 Discussion

Pilot cell difficulty before registering a comparison. Both registered comparisons returned nulls for opposite reasons, position was too hard for every policy and color was too easy for the policies compared. A pilot of a few episodes per cell cannot certify a floor or a ceiling, but it flags them cheaply: a cell that pilots at 0 for 5 is a cell to redesign toward graded difficulty before registering a comparison on it. Our own displacement probe later found the informative region of the position axis between zero and two inches; a pilot would have put the registered comparison there.

The dissociation sits between representation and readout. Position is linearly decodable from every policy, including those whose trajectories ignore it, and a two-inch displacement well inside that decoding accuracy still produces a floor because the sweep never covers it. The representation is frozen and shared; what collection strategy changed is what the action output is conditioned on. That reframes what appearance-axis evaluations measure here: a trajectory-replaying policy passes an unseen-color test by blindness rather than robustness, so the registered ceiling was structural, and it is where we diverge from Xie et al. (2023), who find object texture hard.

Diversity is a purchase with a floor price. Color diversity was nearly free and nearly useless, because appearance does not condition the policies' actions. Position diversity was the consequential axis, and buying it at 50 demonstrations cost execution outright, the trade Belkhale et al. (2023) predict and Xie et al. (2023) observe as training environments grow more varied, here with a specific failure mode: not noisier reaching but no reaching at all in a fifth of episodes. The density probe brackets the price: 25 demonstrations per position restore execution completely and buy no interpolation, so coverage and competence are bought separately; neither split we measured bought both. Sampling Randomized's ten positions at Lin et al. (2024)'s recommended rate would take 500 demonstrations, roughly 2.6 hours of recording at our measured pace, ten times the budget a practitioner at this end of the ecosystem actually spends. The next experiment walks that budget, the same ten positions at 10, 25 and 50 each, to find where diversity stops costing execution and whether the two-mode selector becomes a position map.

Policies inherit the demonstrator, and the outcome is decided late. The demonstrated drop is reproduced at its location while the correction transferred in 6 of 53 releases, and demonstration velocity orders completion time; under a fixed window a slow demonstrator costs throughput before competence. The split is not arbitrary: the release happens at a fixed point of the carry, so it is replayable, while a dropped cube bounces to a different spot each time, so the re-grasp is a position-generalization problem, and it fails for the same reason New Positions does. The policy inherited exactly the half of the behavior that does not require localizing the cube. No early warning of failure was linearly present before the grasp, consistent with most episodes being decided there. SAFE reads outcome from 3B to 7B parameter VLAs; whether the pre-grasp null persists at that scale, or contact physics makes the outcome irreducibly late-breaking, is open.

For practitioners. At a few dozen demonstrations: spend them on the configurations you will deploy, at no fewer than about 25 each, rather than spreading them thin; pilot every evaluation cell before committing to it, since a cell at floor or ceiling cannot resolve anything; and record the trajectories, because the cheapest instrument in this study, commanded joint positions you already log, is the one that explained every result.

Limitations. Appendix I gives the full register. **Power:** at 75 episodes per condition, intervals are roughly ± 14 points; the largest within-condition swing between seeds is 12 points and none is significant, so both registered nulls are inconclusive rather than evidence of absence. **Scope:** one task, arm, model and bench, 50 demonstrations per condition, two seeds; the density probe is one seed. The mechanisms are measured, their generality is not. **Training:** only the action expert is fine-tuned, and every policy executes the full 50-action chunk before re-observing, the least reactive setting in the base model's own ablation; whether position diversity would reshape perception under full fine-tuning, or a shorter horizon would convert near misses into corrections, is untested. **Instruments:** telemetry is commanded rather than achieved position, the release detector is blind within 10 degrees of the grasp, T2 and the cup are angularly indistinguishable from the base so the density probe's T2 cell rests on visual scoring alone, and pace and training exposure were uncontrolled.

7 Conclusion

At a 50-demonstration budget, collection strategy changes trajectories before it changes success rates, and sometimes only there. Repetition produced a policy that executes one motion and perceives nothing; position diversity produced a policy that perceives, cannot execute, sometimes does not move at all, and reached the lowest training loss of the ten; two positions at twenty-five each produced one that executes and selects between its trained behaviors rather than perceiving; color diversity was nearly free and nearly unnecessary; recovery demonstrations taught the failure at its exact location, not the recovery.

Broader impact. Making small-budget adaptation more predictable reduces the hardware trials and compute spent on runs that could not have worked, and the released protocol, telemetry tools and rollout data let other low-cost adapters measure what their policies do rather than only whether they succeeded. The risk runs the other way: cheaper adaptation invites deploying policies validated on success rate alone, which this paper shows can hide both what a policy perceives and what failure it has inherited. A policy that does not look at the scene is safe only while the scene does not change.

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Reproducibility and disclosure Code, data, per-run manifests, the ten training configurations, all 49 evaluation rollout datasets and the released protocol are at `https://github.com/Andresg324/so101-vla-data-study`; every measurement regenerates through `analysis/README.md`. Fine-tuning took roughly 0.75 A100-hours per policy, about 8 in total; collection, evaluation and analysis ran on a laptop, and total hardware and compute cost was roughly $600. SmolVLA, LeRobot and the SO-ARM101 design are used under Apache 2.0; our code is released under the MIT license, and released datasets carry a dataset card describing collection conditions, episode counts and scoring criteria. Drafting and code were AI-assisted; all design decisions, analyses and claims are the author's.

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A Apparatus

Geometry and materials. The overhead camera images a 22×17 inch region of a plywood surface finished in flat gray primer. Positions are reported in inches from the front-left corner of the overhead frame. The cup is 4.5 inches tall with a 3.5 inch rim and is fixed at $(5.0, 12.5)$. The arm is clamped to the front edge, centered at $x = 11.0$. The trained position T6 is $(15.5, 10.0)$. Cubes are 1-inch foam, identical across colors in size, shape, mass and finish. Cameras are a Logitech C270 on the gantry (overhead) and a Seeed webcam at the wrist, both 640×480 at 30 fps. Servos are Feetech STS3215; the leader runs at 5 V and the follower at 12 V. Baseline lighting is two 1000 lm, 4000 K, CRI 80 clamp LEDs bounced off the ceiling, blinds closed and room lights off.

Rollout timing. The policy emits 50 actions per forward pass and executes all 50 before the next, so the action-update rate is 0.600 Hz, one decision every 1.67 s, and playback shows brief holds at chunk boundaries. Frames are recorded only while a chunk executes, not during the forward pass that produces the next one, so a full-length episode is 1151 recorded frames, 38.4 s of motion in 23 chunks inside the 45 s wall-clock window. Every duration computed from recorded data is execution time, not elapsed time. The base model's asynchronous inference stack was not used.

Collection procedure. Both varied conditions cycle their factor rather than blocking it: Randomized runs five passes of the ten marked positions in fixed order, and Color-varied runs ten passes of the five colors. Cycling spreads drift in teleoperator skill evenly across factor levels instead of loading it onto whichever level was collected last, and it makes every episode's factor level recoverable from its index alone, which is what allowed the resumed Randomized session to be verified after the mid-session crash (Amendment 19). Recovery's 20 drop demonstrations occupy slots 2 and 4 of each group of five (Amendment 9), so the drop behavior is likewise not confounded with position in the session.

Surface marks. Position marks for T1 to T10 and E1 to E5 were made before any training demonstration and kept through all rollouts, so they are visible to the policy identically in training and evaluation and across every condition. The displacement probe ran with two additional faint pencil marks at the 1 and 2 inch positions, added after the second evaluation grid closed; those pencil marks were erased before the density probe's collection and evaluation, so the surface matched its prior state.

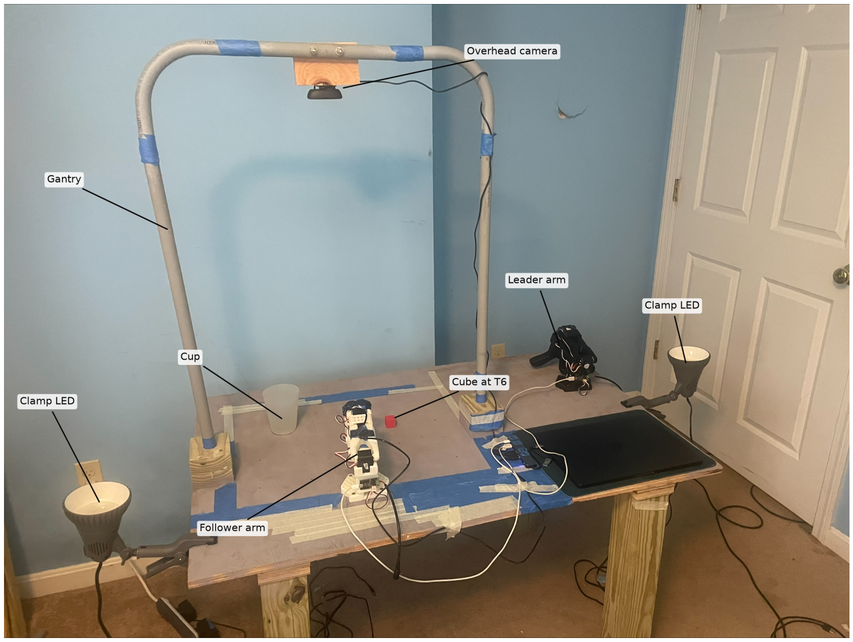

Figure 5: The bench. The overhead camera hangs at the apex of a free-standing conduit arch and images a 22 $\times$ 17 inch gray primed region of the plywood surface. The blue tape marking the board edges lies outside the camera frame; the position markings lie inside it, with coordinates in Figure 1. The camera's own view is Figure 7. Both clamp LEDs bounce off the ceiling for the baseline condition, recorded with blinds closed and room lights off; this photograph was taken under room light, and Figure 9 shows the two lighting conditions as the camera records them.

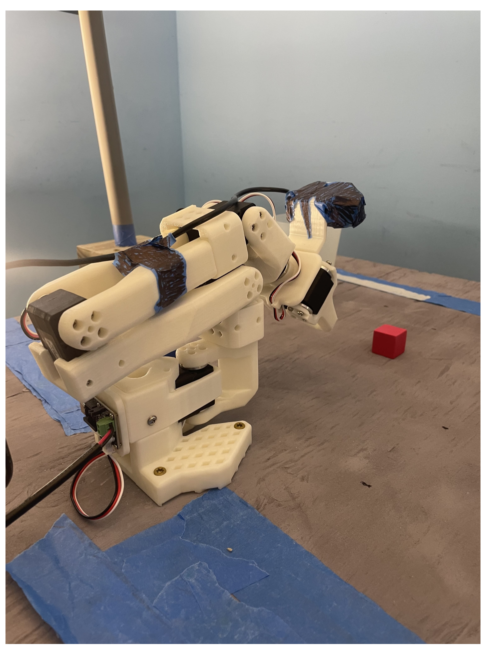

Figure 6: Arm home pose. Every demonstration and rollout begins here, fully retracted with the gripper visible in the overhead frame, and the arm returns to it between episodes.


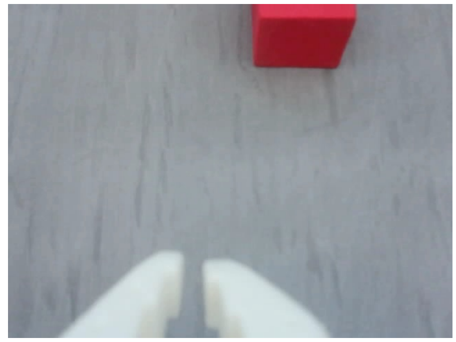

Figure 7: Overhead view (left) and wrist view (right), at the 640×480 resolution the policy receives. These two images are the policy's entire input.

Figure 8: The six cube colors, identical in size and material. Green is held out for evaluation and never appears in training.

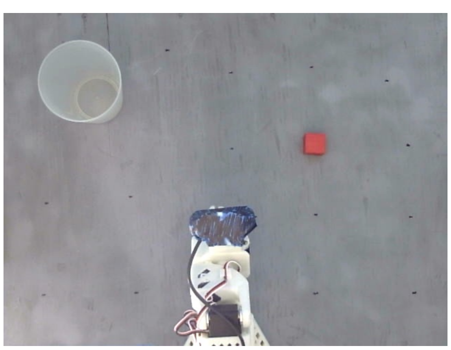


Figure 9: Baseline lighting (left) against reduced lighting (right), as recorded by the overhead camera. Mean grayscale intensity falls from 144.8 to 101.0.

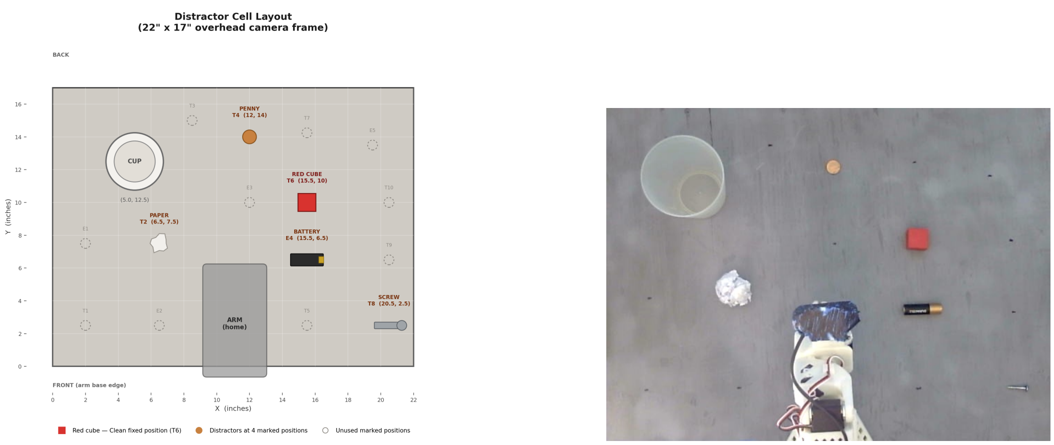

Figure 10: Distractor cell. Left: annotated layout, crumpled paper at T2, penny at T4, battery at E4, screw at T8, cube at T6. Right: as recorded by the overhead camera.

B Training details

All runs use LeRobot's SmolVLA defaults: learning rate 10^{-4} with 1,000 warmup steps, a 50-action chunk, 10 flow-matching sampling steps, expert width multiplier 0.75, batch 32, 10,000 steps, `freeze_vision_encoder=True` and `train_expert_only=True`. The ten resolved configurations differ only in `output_dir`, `seed`, `dataset.repo_id`, `job_name` and `wandb.run_id`. The base model's reference real-world fine-tunes run 200,000 steps at batch 64 and its authors note that fewer suffice (Shukor et al., 2025); every run here plateaus by roughly step 6,000 (Figure 11). One default is disclosed rather than chosen: the cosine schedule is configured for 30,000 decay steps, three times the training run, so the final checkpoint sits at about 79 % of peak learning rate rather than fully annealed, identically for every run. The base model's asynchronous inference stack was not used.

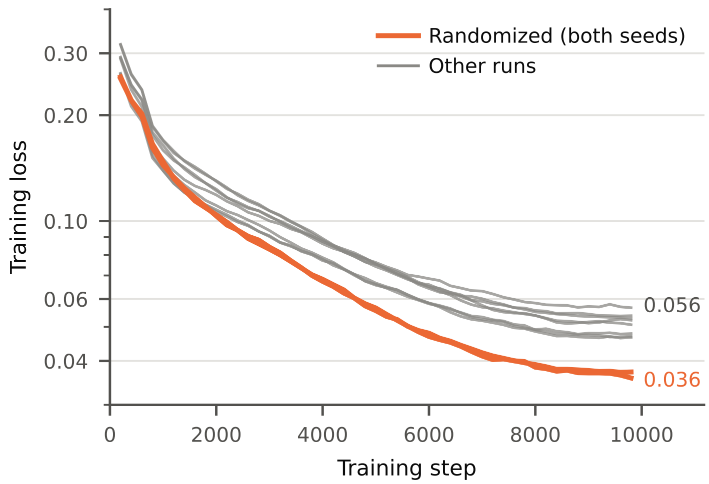

Figure 11: **Training loss does not order the policies the way evaluation does.** All ten fine-tuning runs converge by roughly step 6,000, logged through step 9,800 of 10,000. Randomized reaches the *lowest* final loss of any run, 0.038 and 0.036, while scoring 12/75 and 13/75 at evaluation; the two highest belong to Density, 0.056, perfect at both positions it trained on, and Recovery, 0.054 and 0.053, which scores two to three times better than Randomized.

C Instrument validation

Pan-to-bearing calibration. Base rotation is mapped to workspace bearing by a linear fit on the 50 Randomized training grasps, whose cube positions are known: bearing $= 0.9485\,\text{pan} + 3.669$. Leave-one-out median absolute error is 0.86 degrees (90th percentile 2.22, $R^2 = 0.9990$, $n = 50$). Within-position spread of base rotation is 0.38 to 1.93 degrees at all ten positions, which also confirms the episode-to-position mapping across the resumed Randomized session (Amendment 19). The fit is validated at two independently known locations it never saw: the cube at T6 (true bearing 24.23; Clean's in-distribution grasps land at 23.32 and 24.27 across the two seeds) and the cup (true bearing -25.6; all 362 successful episodes release at a median 2.6 degrees from center, IQR 2.0 to 3.3). The full 2D joint-to-position map is substantially weaker, roughly 1.2 inches lateral and 2.3 inches radial error because grasp height varied across demonstrations, so all quantitative trajectory claims use bearing.

Grasp detector. A grasp is the first sustained gripper closure after the approach opening. An episode whose first opening runs to the last frame, a hover with no close, yields no grasp pose. On the Clean displacement probe this excludes 2 of 8 episodes at one inch and 2 of 8 at two inches from the grasp-pose analysis; the envelope and success figures in Section 5.2 come from the full trajectory and are unaffected. All 50 calibration demonstrations contain a detected grasp.

Release detector. A release is detected when the gripper command opens by at least 12 degrees sustained for 5 frames from its approach-open state. Thresholds were fixed by sweep on labeled demonstration data before any rollout was processed: 20/20 recall at 0/30 false positives on the 20 drop and 30 non-drop demonstrations. On rollouts it recovers 43/43 `deliberate_drop`, 6/6 `success_after_drop` and 6/10 `grasp_drop`, with no false positive across the plain successes. The four misses are travel-filtered rather than undetected openings, the largest azimuth travel among them being 6.6 degrees against a 10 degree filter, and all four were confirmed as genuine drops on video. The detector therefore cannot see drops within 10 degrees of the grasp, which biases release-location estimates toward the cup; the release-bearing estimate moves by less than 2 degrees across the swept settings. Across all episodes with at least one detection, 55 carry a drop-consistent label and 34 do not (20 `contact_no_grasp`, 14 `timeout_other`): openings with no cube in hand, a mode absent from the demonstrations the detector was calibrated on. The release analysis in Section 5.6 is label-filtered, so none of the 34 enter it; all 53 Recovery release rows come from episodes labelled `deliberate_drop` (43), `success_after_drop` (6) or `grasp_drop` (4).

No-departure label. An episode is labeled `no_departure` when the arm never leaves the home pose. The 20 degree departure threshold was chosen by sweeping candidate values against the pre-registered labels across the 662 episodes scored at the time; the 45 density episodes added later reproduce under the same threshold. The two groups separate completely: the largest maximum joint deviation among episodes labelled `no_departure` is 15.9 degrees, and the smallest among episodes labelled as departing is 59.5 degrees. Every threshold from 16 to 59 degrees reproduces the labels exactly, and 20 was taken as the conservative end of that band. One episode initially coded `timeout_other` measured 4.97 degrees, was reviewed on video and recoded; after that correction telemetry and human coding agree on all 707 scored episodes.

Label audit. All 707 scored episodes were screened against two telemetry criteria independent of the label under test: episode duration, since a success terminates when achieved while a failure runs out the window, and the presence of a gripper release inside the cup's angular half-width. All 362 successes contain a cup release; all 27 failures containing one ran the full window, which the success criterion permits. Eight of the 27 are Density episodes on held-out positions: T2 lies at -31.0 degrees, inside the 7.2 degree half-width around the cup at -25.6, so a gripper opening at T2 is indistinguishable from a cup release by bearing. The release screen therefore cannot independently corroborate Density's `trained_t2` cell, which rests on visual scoring alone; the same geometry keeps Density out of the drop table, since a 5.4 degree traverse is below the detector's travel filter. The screens flagged one transposed pair of adjacent episodes (Color, in-distribution, seed 1000, episodes 8 and 9), re-scored from video and corrected, with neighbors 10 and 11 also reviewed and matching. Because the swap exchanges one success for another inside the same cell, no rate, interval or test statistic changed.

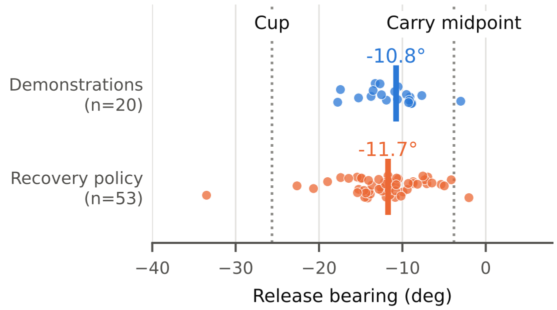

Figure 12: **The policy releases where the demonstrator released.** Bars are medians, points jittered vertically; the cup is at $-25.6°$ and the cube at $+24.2°$, off panel to the right.

Seed 1000 timeout audit. Failure modes at seed 2000 were coded live; seed 1000 codes were assigned retrospectively from contemporaneous notes and video, and `timeout_other` accounts for 57.7 % of seed 1000 failures against 41.3 % at seed 2000. All 94 seed 1000 `timeout_other` episodes were therefore audited. Every one ran to the recording ceiling (median 38.3 s) with substantial joint motion (maximum deviation 92 to 192 degrees), confirming the timeout label in all cases. Eighty-five were re-scored from video, using the wrist camera for the reduced-lighting cells and for episodes with a detected gripper opening; seven were recoded as `contact_no_grasp`. Fourteen episodes carried a detected gripper opening but showed no grasp on video, and were retained as timeouts rather than recoded as drops. All 37 New Positions episodes were confirmed as no-contact timeouts, consistent with the convention that contact was recorded at scoring time. Departure labels agreed with telemetry in all 707 episodes.

Activation extraction. SmolVLA samples action noise, so hidden states are one draw rather than a fixed function of the input: two extractions of the same episode correlate at 0.998 on the same device and 0.997 across CUDA and MPS, so device choice is irrelevant beside the sampling. Extraction is seeded per episode, and the position-probe values reproduce across independent draws to within 0.6 degrees.

D Probe configuration

Extraction. Activations are taken at `model.vlm_with_expert.lm_expert.norm`, the action expert's final normalization and the last representation before the action head (720 dimensions), by replaying recorded rollouts through each policy. This is the same locus SAFE (Gu et al., 2025) reads, the final-layer hidden state before decoding. Replay fidelity was verified by comparing regenerated actions to recorded ones; agreement is within the policy's own sampling noise. One vector is taken per decision chunk, one forward pass every 50 frames (1.67 s), giving about 23 per full-length episode.

Position probe. Ridge regression with cross-validated regularization on the leading 50 principal components of the standardized activation, predicting the cube's true bearing, trained and evaluated on New Positions episodes (approximately 340 to 360 samples per policy). Two split rules are reported, both 5-fold and both grouped: held out by episode, and held out by position, the latter equivalent to leave-one-position-out across the five held-out positions. Chance is the median absolute error of the best constant predictor, the mean bearing, which is 42 degrees. Intervals are cluster bootstraps over the grouping unit that defines the split, 2000 resamples, 2.5 and 97.5 percentiles; the by-position intervals rest on five groups and are correspondingly wide.

Outcome probe. Logistic regression (L2, $C = 0.1$) on the leading 10 principal components, predicting binary success from one vector per episode: the mean activation over a three-chunk window ending at an information cutoff, swept in chunk-size steps across the episode. One vector per episode rather than per frame, so episode duration cannot leak, since successes terminate early and failures run the full window. Evaluation is 5-fold stratified AUC. Significance is assessed against a permutation null in which labels are shuffled within each evaluation cell, 500 permutations per policy,

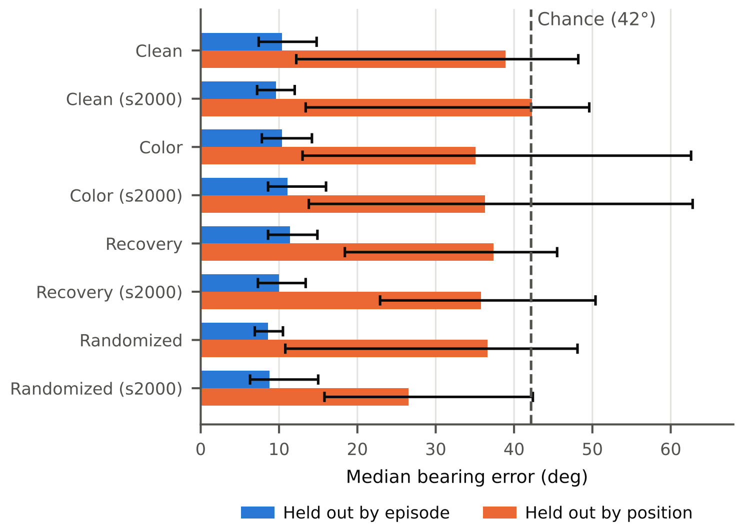

Figure 13: Median bearing decoding error for all eight policies under both split rules, with 95 % intervals and the chance floor. Position is decodable in every policy; holding out whole positions collapses it to chance.

because cells differ in base rate and appearance and a probe could otherwise score above chance by recognizing the cell alone; the reported band is that null's 95th percentile, which sits at 0.64 to 0.74 rather than 0.5. Cells with no outcome variation are dropped. A control probe on the last three chunks reaches AUC 0.99 to 1.00 for every policy.

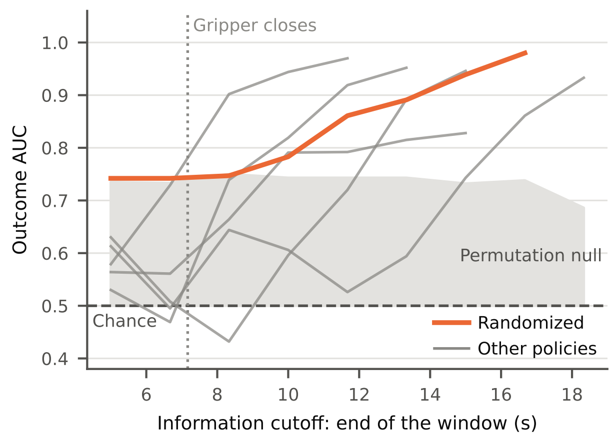

Figure 14: **The outcome is unreadable until the gripper closes.** Outcome-probe AUC against the end of its five second window; the shaded band is the permutation null's 95th percentile.

Inclusion. Cells with outcome variance are pooled per policy: Clean seed 2000 (different object, distractors, reduced lighting and the one-inch displacement probe; 34 successes, 19 failures), Color seed 1000 (distractors, in-distribution, reduced lighting; 38/7), Randomized seeds 1000 and 2000 (12/33 and 13/32), Recovery seeds 1000 and 2000 (39/21 and 31/29). Clean seed 1000 (5 failures) and Color seed 2000 (4) fall below a pre-stated floor of six episodes in the rarer outcome and are excluded, since AUC and stratified folds cannot be supported below it. Because successful episodes terminate early, sample size and class balance shift at long cutoffs; the sweep is truncated at the last offset before any episode drops out.

E Registered grid, matched comparisons and generalization gap

Table 2: Matched-axis comparisons against Clean. Both are null: one because no policy succeeded at all, one because both policies were at ceiling.

Cell	Policy	Score	Clean	Difference	95 % CI	Fisher p
New positions	Randomized (s1000)	0/15	0/15	0.000	-0.204, 0.204	1.000
New positions	Randomized (s2000)	0/15	0/15	0.000	-0.204, 0.204	1.000
Different object	Color (s1000)	15/15	15/15	0.000	-0.204, 0.204	1.000
Different object	Color (s2000)	15/15	12/15	$+0.200$	-0.042, 0.452	0.224

Table 3: Success rates with Wilson 95 % intervals, seed 1000.

Condition	Cell	Successes	Rate	CI low	CI high
Clean	in-distribution	15/15	1.000	0.796	1.000
Clean	new positions	0/15	0.000	0.000	0.204
Clean	reduced lighting	12/15	0.800	0.548	0.930
Clean	different object	15/15	1.000	0.796	1.000
Clean	distractors	13/15	0.867	0.621	0.963
Color	in-distribution	13/15	0.867	0.621	0.963
Color	new positions	0/15	0.000	0.000	0.204
Color	reduced lighting	12/15	0.800	0.548	0.930
Color	different object	15/15	1.000	0.796	1.000
Color	distractors	13/15	0.867	0.621	0.963
Recovery	in-distribution	8/15	0.533	0.301	0.752
Recovery	new positions	0/15	0.000	0.000	0.204
Recovery	reduced lighting	7/15	0.467	0.248	0.699
Recovery	different object	11/15	0.733	0.480	0.891
Recovery	distractors	13/15	0.867	0.621	0.963
Randomized	in-distribution	6/15	0.400	0.198	0.643
Randomized	new positions	0/15	0.000	0.000	0.204
Randomized	reduced lighting	5/15	0.333	0.152	0.583
Randomized	different object	0/15	0.000	0.000	0.204
Randomized	distractors	1/15	0.067	0.012	0.298

The gap is bounded above by the in-distribution rate, so Clean is penalized for being perfect in distribution and Randomized at 0.400 cannot post a gap above 0.400 however completely it collapses. The ranking is unstable across seeds for the same reason. Recovery leads at seed 1000 because both of its rates sit near 0.5, and Color leads at seed 2000 through the highest held-out mean in the study combined with a lower in-distribution starting point than Clean.

F Displacement probe, rollout motion and training loss

Table 4: Success rates with Wilson 95 % intervals, seed 2000.

Condition	Cell	Successes	Rate	CI low	CI high
Clean	in-distribution	15/15	1.000	0.796	1.000
Clean	new positions	0/15	0.000	0.000	0.204
Clean	reduced lighting	9/15	0.600	0.357	0.802
Clean	different object	12/15	0.800	0.548	0.930
Clean	distractors	10/15	0.667	0.417	0.848
Color	in-distribution	12/15	0.800	0.548	0.930
Color	new positions	0/15	0.000	0.000	0.204
Color	reduced lighting	14/15	0.933	0.702	0.988
Color	different object	15/15	1.000	0.796	1.000
Color	distractors	15/15	1.000	0.796	1.000
Recovery	in-distribution	9/15	0.600	0.357	0.802
Recovery	new positions	0/15	0.000	0.000	0.204
Recovery	reduced lighting	5/15	0.333	0.152	0.583
Recovery	different object	5/15	0.333	0.152	0.583
Recovery	distractors	12/15	0.800	0.548	0.930
Randomized	in-distribution	6/15	0.400	0.198	0.643
Randomized	new positions	0/15	0.000	0.000	0.204
Randomized	reduced lighting	5/15	0.333	0.152	0.583
Randomized	different object	2/15	0.133	0.037	0.379
Randomized	distractors	0/15	0.000	0.000	0.204

Table 5: Generalization gap by condition and seed, ranked smallest to largest. The gap is bounded above by the in-distribution rate, so it measures the drop from a policy's own baseline and not how well the policy performs; the held-out mean is the interpretable column. Every held-out mean includes New Positions at 0/15.

Condition	Seed	In-distribution	Held-out mean	Gap
Recovery	1000	0.533	0.517	0.017
Color	2000	0.800	0.733	0.067
Color	1000	0.867	0.667	0.200
Recovery	2000	0.600	0.367	0.233
Randomized	2000	0.400	0.117	0.283
Randomized	1000	0.400	0.100	0.300
Clean	1000	1.000	0.667	0.333
Clean	2000	1.000	0.517	0.483

Table 6: Displacement probe, Clean at both seeds, 8 episodes per point. Outcomes shift from contact-without-grasp at one inch to no-contact timeouts at two.

Displacement	Seed	Success	Contact, no grasp	Timeout	No departure	Sweep reached cube
0 in (T6)	1000	15/15	0	0	0	15/15
0 in (T6)	2000	15/15	0	0	0	15/15
1 in	1000	0/8	6	1	1	6/8
1 in	2000	3/8	5	0	0	4/8
2 in	1000	0/8	0	7	1	0/8
2 in	2000	0/8	1	7	0	0/8

Table 7: Final training loss against evaluation score for all ten runs, sorted by loss. Loss is the last logged value at step 9,800.

Run	Final loss	Grid score
Randomized (s2000)	0.0357	13/75
Randomized (s1000)	0.0375	12/75
Color-slowpace (s1000)	0.0472	24/30 (2 cells)
Clean (s2000)	0.0473	46/75
Clean (s1000)	0.0482	55/75
Color (s2000)	0.0508	56/75
Color (s1000)	0.0518	53/75
Recovery (s2000)	0.0529	31/75
Recovery (s1000)	0.0538	39/75
Density (s1000)	0.0559	30/30 trained, 0/15 held out

Table 8: Rollout motion by condition and seed. Latency is median time to first departure; deg/step and duration are over successes. Clean pools the 16 displacement-probe episodes per seed; Density pools its three cells.

Condition	Seed	Never departed	Latency (s)	Deg/step	Duration (s)
Clean	1000	4/91	6.63	0.776	19.6
Clean	2000	7/91	6.88	0.778	21.5
Color	1000	0/75	1.03	0.887	13.4
Color	2000	0/75	1.57	0.887	13.5
Recovery	1000	3/75	3.21	0.915	17.7
Recovery	2000	5/75	2.15	0.867	21.1
Randomized	1000	11/75	5.57	0.899	21.6
Randomized	2000	18/75	8.13	0.903	23.1
Color-slowpace	1000	2/30	5.17	0.829	22.4
Density	1000	0/45	0.63	0.943	13.4

G Failure-mode vocabulary

Episodes are labeled by their first decisive event: a cube that was held and then released is `grasp_drop` wherever it comes to rest, and `cube_out_of_bounds` is reserved for episodes in which the cube leaves the workspace without ever having been held. Counts are over all 707 scored episodes including the 45 Density rollouts; the three success labels sum to 362 successes.

Table 9: Outcome vocabulary and counts.

Label	Count	Definition
`success`	350	Cube released and resting in the upright cup, first attempt
`success_after_missed_grasp`	6	Initial grasp failed; policy re-approached and completed
`success_after_drop`	6	Grasped, released mid-carry, recovered, task completed
`timeout_other`	171	Window expired without success or another decisive event
`contact_no_grasp`	70	Cube contacted and displaced without a grasp; not re-acquired
`no_departure`	50	Arm never left the home pose (validated, Appendix C)
`deliberate_drop`	43	Mid-carry release matching the imitated Recovery behavior
`grasp_drop`	10	Cube held and lost without the deliberate mid-carry signature
`cube_out_of_bounds`	1	Cube left the workspace without ever having been held
`cup_knocked`	0	Cup knocked over (immediate failure)

H Amendment log

The protocol was amended 30 times, each amendment dated and stating which data it precedes. Full text is in the released protocol; this table condenses them.

Table 10: Amendments, condensed. 1 to 13 precede all data collection, 14 to 17 fall between the two seed grids, 18 to 29 are analysis stage, 30 precedes the follow-up policy in Section 5.4.

#	Stage	Summary
1	pre-collection	Workspace geometry: 10 training and 5 held-out positions
2	pre-collection	Color condition rebalanced to five colors $\times$ 10
3	pre-collection	Background changed from blue tape to gray primer
4	pre-collection	Lighting cell reduced to a single level (one LED off)
5	pre-collection	Scoring changed to live scoring with all video retained
6	pre-collection	Seeds: commit to reporting the number run; seed fixed at 1000
7	pre-collection	Recovery given an operational definition (drop point, height)
8	pre-collection	Demo ordering specified as cycling rather than blocking
9	pre-collection	Recovery demos placed at fixed slots (2 and 4 of each 5)
10	pre-collection	Training hyperparameters, seed and checkpoint selection fixed
11	pre-collection	Recording window and language instruction fixed
12	pre-collection	Matched comparisons: Newcombe interval and Fisher test
13	pre-collection	16th warmup episode per cell, discarded unconditionally
14	between grids	Seed 2000 replication declared; fixed cell order
15	between grids	Displacement probe declared exploratory
16	between grids	Pace probe declared exploratory
17	between grids	Failure-mode vocabulary fixed (superseded by 21 and 28)
18	analysis	Color re-collection documented; superseded dataset retained
19	analysis	Randomized crash and resume; index-to-position verified
20	analysis	Re-record rules stated for demonstrations and evaluation
21	analysis	Vocabulary corrected: the two `success_after` labels
22	analysis	Instrument validation of live labels (Appendix C)
23	analysis	Bearing analyses declared exploratory; commanded-position basis
24	analysis	Correction: drop point is $\sim$65 % of the carry, not the midpoint
25	analysis	Correction: action-update rate is 0.600 Hz, not $\sim$3 Hz
26	analysis	One outcome-dependent evaluation re-record logged
27	analysis	Label audit; one transposed pair corrected
28	analysis	Vocabulary precedence rule: first decisive event
29	analysis	Training configuration verified from resolved configs; frozen backbone, LR schedule
30	follow-up	Density probe: 25 demonstrations at T6 and 25 at T2, single seed, three cells, declared before collection

I Limitations, full register

The limitations paragraph in Section 6 states the bounds on the headline claims. This is the complete list.

Power and seeds. At 15 episodes per cell and 75 per condition only large differences are detectable: Wilson intervals are roughly ± 14 points at $n = 75$ and closer to ± 25 at $n = 15$, and across the two seeds no within-condition difference is significant, so a difference under about 13 points is unresolvable whatever its source. Nulls are reported as inconclusive, not as evidence of no effect. The two replications are reported separately, never pooled, and bound training-run variance only loosely.

Partial fine-tuning. Fine-tuning modifies only the action expert; the vision encoder and language backbone are frozen, so roughly 22 % of the model is trained. Whether position diversity would reshape perception under full fine-tuning, rather than only the action readout, is untested. The position code the probe reads was learned in pretraining and is shared by every policy by construction.

Open-loop chunk execution. All policies execute the full 50-action chunk before re-observing. In the base model's own ablation this is the least reactive setting, costing roughly 29 points of success against re-observing every step (Shukor et al., 2025). The fixed sweep is therefore measured under a configuration giving the policy 23 opportunities to re-observe across an episode rather than 1151, and a shorter execution horizon might convert some near misses into corrections.

No augmentation. No image augmentation was used. Xie et al. (2023) find random-crop augmentation cheaply improves robustness to both spatial and non-spatial factors; whether it would move the one-inch boundary is untested.

Uncontrolled covariates. Because the budget is fixed in episodes rather than frames, conditions receive between 9.9 and 12.9 passes over their own data at a fixed 10,000 steps. Demonstration pace was likewise uncontrolled and spans 16.6 to 21.4 s per demonstration; the pace probe suggests pace is not inert, and no claim here separates pace from the manipulated factor. The default cosine schedule decays over 30,000 steps, so the final checkpoint is not fully annealed; every run's loss plateaus by step 6,000 regardless.

Instrument limits. All telemetry derives from commanded rather than achieved joint positions, so gripper telemetry cannot distinguish a closure blocked by the cube from a closure on empty air; claims about whether the cube was held rest on live scoring and retained video. The release detector cannot see drops within 10 degrees of the grasp, biasing release-location estimates toward the cup; it under-counts rather than over-counts. On rollouts it also fires on openings with no cube in hand; these are excluded from the release analysis by label (Appendix C).

Density probe. Single seed, post hoc, 45 rollouts, recorded in a separate session two weeks after the grid by the same operator, by then more practiced (17.6 s per demonstration, inside the registered range). T2 and the cup are 5.4 degrees apart from the base, so the telemetry screens that corroborate every other cell cannot corroborate the T2 cell (Appendix C), and E3 lies about one inch off the T2–T6 chord, so its interpolation read is weaker than the two trained-position cells.

Partial observability. At four of the ten Randomized training positions (T1, T8, T9, T10) the gripper leaves the overhead frame during part of the approach while the wrist camera stays on target. This affects only the Randomized condition and follows from the fixed camera geometry.

Adversarial distractors. Distractors sit at marked cube positions rather than random ones, making that cell adversarial by construction and adversarial to a different degree for Randomized, three of whose training positions are occupied.

Single operator, unblinded scoring. One experimenter demonstrated, evaluated and scored; the demonstrator and evaluator roles Kress-Gazit et al. (2024) recommend separating were not, and scoring was not blind to condition. The criterion is binary and physically unambiguous, was fixed before collection, cells ran in a pre-registered order, all video is retained, ambiguous episodes were re-scored from video, and a post-hoc telemetry audit of all 707 episodes found one transposed label pair, corrected without changing any statistic. The single demonstrator also controls out operator heterogeneity, so the tempo and release-point findings are attributable to one demonstrator's properties.

Modest extrapolation. The held-out positions lie a few inches beyond the training envelope, within reach and frame; the interpolation against extrapolation split rests on the 12 episodes with a detected grasp and is descriptive rather than a formal test.

Scale. SAFE (Gu et al., 2025) reads success and failure from 3B to 7B parameter VLAs; whether the pre-grasp null reported here persists at that scale is open, with a concrete reason to expect richer features.

Stochastic activations. Probe results are computed on one seeded extraction draw. Independent draws correlate at 0.998 and the reported position-probe values move by less than 0.6 degrees, but no probe number here is exactly reproducible without the seed.

The full limitation register, including items that affect reproduction rather than interpretation, is maintained in the released protocol.